Interest Rates and Exchange Rates: Crisis-Driven Dynamics

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Abstract

We examine the exchange rate response to interest rate surprises using weekly local projections instrumented with high-frequency surprises (LP-IV) for the US, Euro Area, and UK. The weekly frequency preserves identification while mitigating temporal aggregation bias. We provide a decomposition of LP-IV impulse responses that expresses estimated effects as weighted averages of historical outcomes. While baseline estimates indicate immediate exchange rate appreciation following monetary contractions, the decomposition reveals that these responses are driven primarily by a small number of crisis episodes, most notably the Global Financial Crisis, underscoring pronounced state dependence in exchange rate dynamics.

JEL: E52, F31, C36

Keywords: Exchange rate overshooting, Monetary policy surprises, Local projections, High-frequency identification, State dependence, Crisis dynamics

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1 Introduction

The exchange rate overshooting hypothesis, introduced by Dornbusch (1976), predicts that with sticky prices a monetary expansion leads to an immediate depreciation of the exchange rate that exceeds its long-run level, followed by gradual appreciation. Empirical support for this mechanism remains mixed. Early VAR-based studies typically reported delayed overshooting, with peak exchange rate responses occurring several months after a monetary shock (Eichenbaum & Evans, 1995). In contrast, more recent work using high-frequency identification (HFI) of monetary policy surprises often finds rapid exchange rate adjustments, even no overshooting, particularly in advanced economies.

This paper re-examines exchange rate dynamics following monetary policy surprises by combining HFI surprises with weekly local projections estimated by instrumental variables (LP-IV).[1] This framework allows us to trace exchange rate responses at weekly horizons up to one year, preserving the exogeneity advantages of HFI while mitigating temporal aggregation bias inherent in monthly data.

Our contributions are threefold. First, we extend the LP decomposition approach of Goulet Coulombe & Klieber (2025) to an instrumental variables setting, expressing LP-IV estimates as weighted averages of historical realizations. This decomposition reveals that estimated exchange rate responses are heavily influenced by a small number of crisis episodes, most notably during the Global Financial Crisis (GFC). Second, we employ weekly data to better capture the high-frequency transmission of monetary policy surprises. Third, we construct a harmonized dataset for the US, Euro Area, and UK, explicitly accounting for synchronized monetary actions and global financial conditions across advanced economies.

2 Data

We construct a weekly dataset spanning 1999–2025 for the US, Euro Area, and UK. Observations are sampled at the end of each trading week (Friday or the last trading day) to mitigate asynchronous trading effects across time zones.[2]

The analysis focuses on bilateral exchange rates and corresponding short-term interest rate differentials. Specifically, we examine the currency pairs USD/EUR, USD/GBP, and EUR/GBP. Exchange rates are expressed in logarithms and are defined as units of the base (domestic) currency per one unit of the quoted (foreign) currency. An increase in the exchange rate, therefore, corresponds to a depreciation of the base currency. For each currency pair, we compute weekly changes in one-year interest rate differentials, defined as the difference between domestic and foreign swap yields. This aligns with uncovered interest parity, which links ex-

[1] Related studies using monetary policy surprises include Faust *et al.* (2007), Hausman & Wongswan (2011), Rogers *et al.* (2018), Müller *et al.* (2019), Rüth (2020), Gürkaynak *et al.* (2021), Ferrari *et al.* (2021), Cesa-Bianchi & Sokol (2022), Gründler *et al.* (2023), and Yun (2025), which generally document prompt exchange rate appreciation following monetary contractions.

[2] Aggregating to end-of-week also reduces microstructure noise in end-of-day FX quotes.

change rate dynamics to relative monetary stances rather than to movements in a single policy rate.

Monetary policy surprises are obtained from HFI databases: the USMPD from the Federal Reserve Bank of San Francisco (Acosta *et al.*, 2025), the EA-MPD from the European Central Bank (Altavilla *et al.*, 2019), and the UKMPD from the Bank of England (Braun *et al.*, 2025). These datasets provide intraday yield surprises around policy announcements, isolating the unexpected component of monetary policy. US monetary surprises are identified using changes in the fourth-quarter interest rate Eurodollar futures. From January 2022 onwards, we transition to SOFR futures. Similarly, for the UK, we measure surprises in the fourth quarter futures. For the pre-2022 sample, we utilize 3-month LIBOR futures; for the subsequent period, we switch to SONIA futures. For the Euro Area, we identify policy shocks using the change in the 1-year OIS rate. We measure these changes within a monetary event window for the US and EA. For each country, we aggregate announcement-day surprises to weekly frequency by summing all surprises within a given week. To instrument weekly changes in interest rate differentials, we construct pair-specific monetary policy shock differentials by subtracting the foreign shock from the domestic shock.

Control variables include lagged exchange rates, the ICE BofA U.S. High Yield spread, the VIX index, and WTI oil prices, capturing global financial conditions and time-varying risk premia. All data were obtained from LSEG.

3 Empirical Methodology

To examine the impact of monetary policy surprises on exchange rates, we estimate impulse response functions (IRFs) using the LP-IV approach of Jordà *et al.* (2015). This method traces dynamic responses by estimating a sequence of horizon-specific regressions. For each horizon h, we estimate:

$$e_{t+h} = \alpha_h + \beta_h \Delta d_t + \gamma_h(L) W_{t-1} + \text{trend}_t + \epsilon_{t+h} \tag{1}$$

where e_{t+h} represents the log of exchange rate at time $t + h$. α_h and ϵ_{t+h} denote intercept and idiosyncratic error term. The term $\Delta d_t = \Delta i_t^{dom} - \Delta i_t^{for}$ is the weekly change in the one-year interest rate differential, instrumented by the corresponding differential in high-frequency monetary policy surprises $z_t = z_t^{dom} - z_t^{for}$.

The coefficients of interest, β_h, capture the dynamic response of the exchange rate to a relative monetary policy tightening in the domestic economy. A negative value of β_h therefore indicates an appreciation of the base currency. Horizons $h = 0, 1, \ldots, 52$ correspond to responses up to one year. W_{t-1} and $\gamma_h(L)$ represent additional control variables and their estimated coefficients, consisting of lagged dependent variables and other controls. Following Montiel Olea & Plagborg-Møller (2021) and Herbst & Johannsen (2024), we include lagged dependent variables to ensure robust inference. Since lagged endogenous variables are included, IRFs should be interpreted as cumulative (Jordà, 2023).

Instrument strength is assessed using the effective first-stage F-statistic of Olea & Pflueger

(2013). Although the traditional F-statistic threshold of 10 is often used, recent literature advocates for more stringent cutoffs to effectively bound relative bias. To ensure valid inference even under weak identification, we construct confidence intervals using inversion of the Anderson–Rubin statistic (Andrews *et al.*, 2019), which provides correct coverage regardless of instrument strength. For comparison, we also report standard 2SLS confidence bands computed with HAC standard errors.

LP-IV estimates represent average responses over the full sample. To identify the drivers of these averages, we extend the influence-function decomposition of Goulet Coulombe & Klieber (2025) to the just-identified IV setting. The IV estimator $\hat{\beta}_h^{IV}$ is the ratio of the reduced-form (RF) coefficient to the first-stage (FS) coefficient:

$$\hat{\beta}_h^{IV} = \frac{\hat{\pi}_h^{RF}}{\hat{\delta}^{FS}} \tag{2}$$

where the reduced form is $\tilde{e}_{t+h} = \pi_h \tilde{z}_t + \epsilon_{t+h}$ and the first stage is $\tilde{d}_t = \delta \tilde{z}_t + \eta_t$. Variables with tildes denote residuals after partialling out controls.

The reduced-form coefficient can be expressed as a weighted sum of observation-specific outcomes: $\hat{\pi}_h^{RF} = \sum_{t=1}^{T} \omega_t \tilde{e}_{t+h}$. The weights ω_t are determined by the leverage of the instrument: $\omega_t = \frac{\tilde{z}_t}{\sum_{j=1}^{T} \tilde{z}_j^2}$. Substituting into the IV estimator yields the decomposition:

$$\hat{\beta}_h^{IV} = \sum_{t=1}^{T} \underbrace{\left(\frac{1}{\hat{\delta}^{FS}}\right)}_{\text{Scale}} \underbrace{\left(\frac{\tilde{z}_t}{\sum \tilde{z}^2}\right)}_{\text{Shock Intensity}} \underbrace{\tilde{e}_{t+h}}_{\text{Outcome}} \tag{3}$$

The decomposition in Equation 3 decomposes the LP-IV coefficient into three components: Scale (first-stage strength), Shock Intensity (the instrument's leverage), and Outcome (the realization of the exchange rate). This allows us to identify whether a specific result is a general feature of the sample or an artifact of specific high-volatility dates. The decomposition in Equation 3 treats the first-stage coefficient, $\hat{\delta}^{FS}$, as a single global constant. This is consistent with the standard LP-IV framework, which assumes that the pass-through from monetary policy surprises to interest rate differentials is structurally stable over the sample. We acknowledge that in practice, the first-stage relationship may exhibit time variation. However, because our primary goal is to provide a mechanical decomposition of the estimated full-sample LP-IV coefficient, which by definition relies on an average first-stage effect, equation 3 remains the exact arithmetic identity of the baseline result.

To further support our findings and formally address the state-dependent nature of the exchange rate response, the Online Appendix provides results from an interacted LP-IV specification and a multi-horizon rolling window regression analysis. We also verify that the results are robust to varying the number of lags and the trend specification and provide additional details on individual contributions.

4 Results

Figure 1 (top-left panel) reports the impulse response of the USD/EUR exchange rate to a 100-basis-point increase in the bilateral one-year interest rate differential. The estimated response is economically large and statistically significant on impact: a tightening of this magnitude is associated with an immediate appreciation of close to 10 percent. The response remains broadly stable over the projection horizon and persists at the one-year mark, although confidence intervals widen substantially.

The decomposition in Figure 1 (top-right panel) reveals a pronounced heterogeneity in the contributions underlying this average response. The figure plots the cumulative influence of individual weeks on the estimated coefficients at horizons $h = \{0, 4, 8, 12, 24\}$. The aggregate impulse response is largely driven by a small cluster of observations during the GFC and the European sovereign debt crisis. These crisis-induced contributions are followed by sharp reversals once financial conditions normalize, indicating that the large exchange rate responses observed during stress episodes are not persistent. From 2013 to 2020, the cumulative contribution shows a gradual negative drift, indicating weaker exchange rate responses outside crisis periods, and even slight depreciation after 2020. This reflects a lower transmission regime rather than a reversal of earlier effects. These patterns suggest that the estimated average impulse response reflects a mixture of distinct regimes rather than a uniform dynamic pattern.

The bottom panels of Figure 1 show the IV scaling mechanism. The first-stage pass-through of monetary policy surprises to the one-year interest rate differential is approximately 0.6, implying a scaling factor of about 1.6. Nonetheless, the instrument remains strong: the effective first-stage F-statistic exceeds 30, comfortably above conventional weak-instrument thresholds.[3]

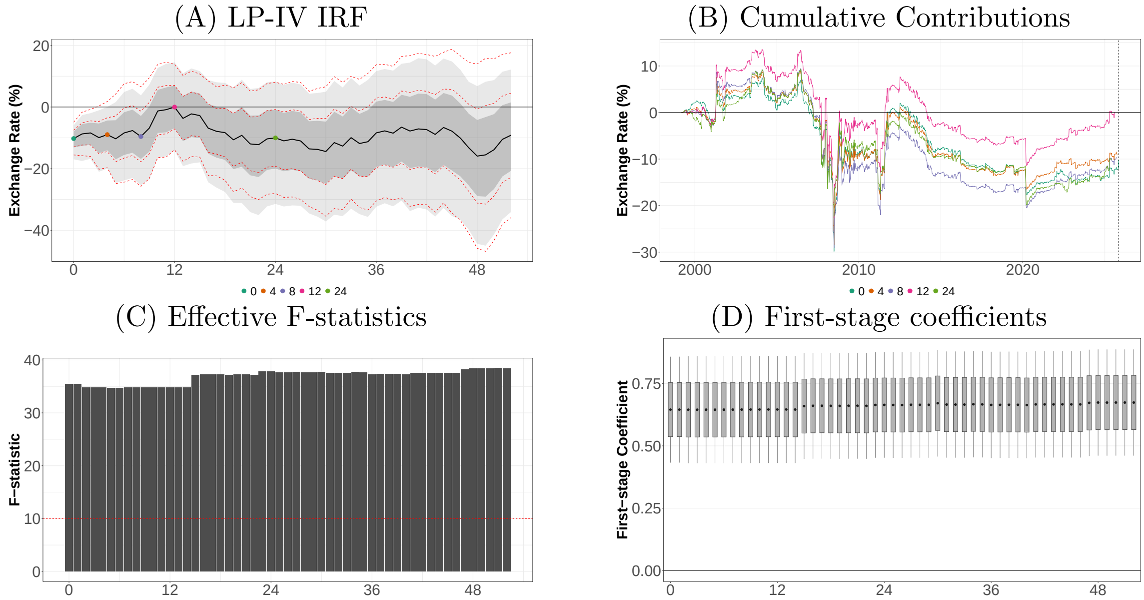
Figure 2 presents results for the USD/GBP exchange rate. The top-left panel shows the impulse response to a 100-bp widening in the one-year bilateral interest rate differential. Dynamics are broadly similar to the euro area: the currency appreciates on impact, with slightly larger magnitude and some delayed overshooting, remaining significant beyond one year.

The decomposition in the top-right panel highlights distinct sources of heterogeneity. While the GFC continues to be a primary driver of the aggregate response, the influence-function analysis also identifies sizable contributions from idiosyncratic UK-specific episodes, most notably the Brexit referendum period and the post-COVID monetary tightening cycle. Pre-2008 contributions often have the opposite sign, reflecting depreciation rather than appreciation. These patterns underscore the strong state dependence of the average response. The bottom panels illustrate IV scaling. The first-stage pass-through is roughly 0.5, amplifying the reduced-form effect, while the instrument remains strong.

Finally, Figure 3 presents the EUR/GBP results. By construction, these responses should mirror those for USD/EUR and USD/GBP, consistent with triangular arbitrage. The estimated dynamics are robust and of comparable magnitude to the previous cases. As before, the aggre-

[3] Because the horizon-specific regressions ($h = 0, 1, ...$) include shifting observation windows and lagged controls, the sample and the residualized regressors vary for each horizon, causing minor fluctuations in the first-stage estimates.

Figure 1: USD/EUR

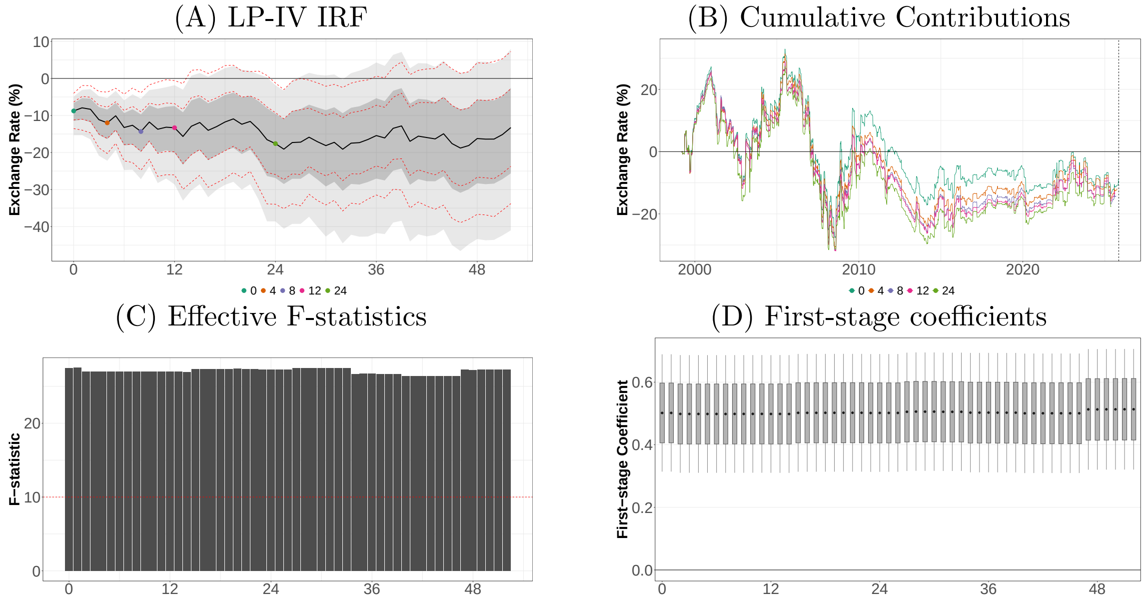


Note: (A) IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% AR CI; red dotted: HAC CI). (B) Week-level contributions to the IRF. (C) Effective first-stage F-statistics by horizon. (D) First-stage coefficients with 16–84% / 5–95% CI.

gate response is largely driven by a small set of high-stress episodes, including the GFC, the European sovereign debt crisis, the dot-com bubble, and Brexit.

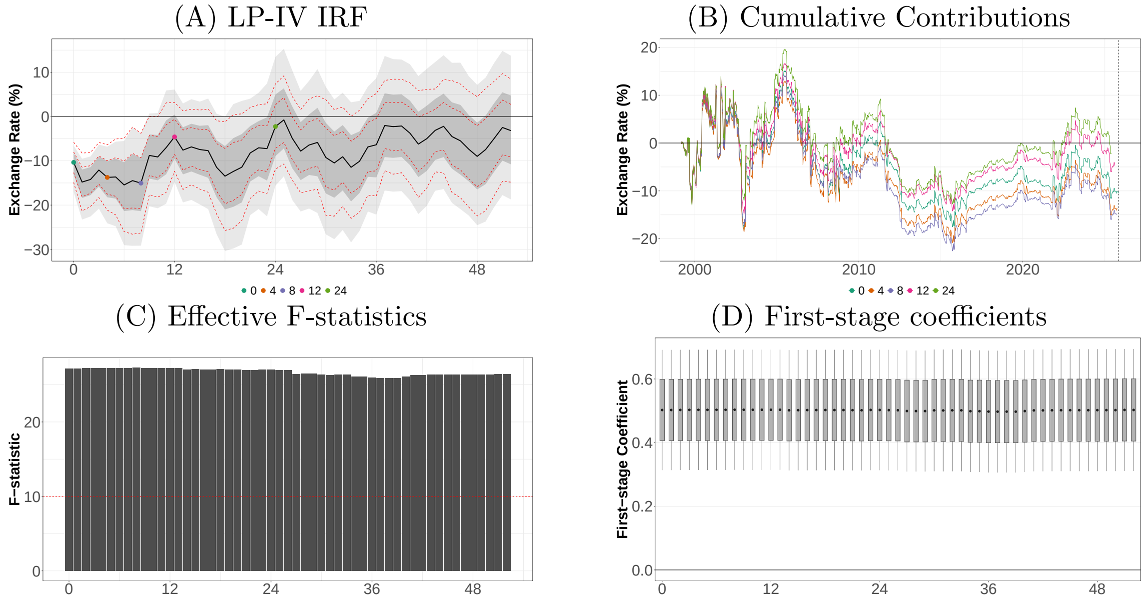
Taken together, our results show that even when monetary policy is measured using bilateral interest rate differentials and identified through high-frequency surprises, exchange rate impulse responses are dominated by a limited number of crisis weeks, with little evidence of systematic overshooting. The dominance of crisis episodes persists despite controlling for global risk premia (VIX, high-yield spreads), suggesting that crises alter the informational weight of monetary surprises rather than just increasing market volatility. As a result, impulse responses obtained from HFI-based approaches may overstate the typical exchange rate reaction to monetary policy surprises outside of crisis periods.

Figure 2: USD/GBP



Note: (A) IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% AR CI; red dotted: HAC CI). (B) Week-level contributions to the IRF. (C) Effective first-stage F-statistics by horizon. (D) First-stage coefficients with 16–84% / 5–95% CI.

Figure 3: EUR/GBP



Note: (A) IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% AR CI; red dotted: HAC CI). (B) Week-level contributions to the IRF. (C) Effective first-stage F-statistics by horizon. (D) First-stage coefficients with 16–84% / 5–95% CI.

5 Conclusion

This paper revisit the exchange rate overshooting puzzle by combining high-frequency identification with a weekly LP-IV framework. Our estimates confirm the prompt currency appreciation following monetary contractions documented in recent literature. However, our decomposition analysis offers a more nuanced perspective: the aggregate impulse responses are largely driven by a small set of high-leverage crisis episodes, most prominently the GFC. This suggests that the exchange rate transmission mechanism is regime-dependent rather than structurally invariant.

This state dependence provides a structural explanation for the persistent failure of exchange rate forecasting models: if transmission shifts radically during crises, models relying on sample averages will necessarily fail to generalize. Furthermore, our findings reconcile the discrepancy between traditional VAR evidence and modern HFI studies. The rapid adjustment often found in HFI research may be an artifact of extreme market sensitivity during specific stress periods rather than a general property of monetary policy.

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Online Appendix

This Online Appendix provides supplementary diagnostics, technical details of the state-dependent specifications, and robustness checks supporting the baseline results presented in the main text.

A Data and Influence Diagnostics

This section describes the properties of our high-frequency dataset and provides diagnostics on the observations driving our central estimates.

Table A3 provides the summary statistics for the interest rate differential changes and the identified monetary shock differentials. At a weekly frequency, the typical magnitude of these innovations is relatively modest. For the USD/EUR pair, for example, the standard deviation of the interest rate differential change is 8.5 basis points, while the monetary shock differential exhibits a standard deviation of only 3.2 basis points. Consequently, while we follow the standard convention of normalizing our impulse response functions to a 100-basis-point contraction for ease of interpretation, it should be noted that such a magnitude represents a significant tail event relative to the typical weekly variation documented in our sample. The reported IRFs thus reflect estimated structural elasticities rather than the impact of a "typical" weekly shock.

To further assess the drivers of our aggregate estimates, we provide a chronological tabulation of the observations exerting the greatest influence on the estimated impulse responses at horizon $h = 0$ (Table A1). These high-leverage episodes are concentrated in periods of heightened financial stress, most notably around the Global Financial Crisis and Eurozone sovereign debt crisis, and account for a disproportionate share of the aggregate response. Earlier observations, primarily from the pre-2008 period, often contribute with the opposite sign, implying exchange rate depreciation rather than appreciation. This divergence suggests the presence of distinct monetary and financial regimes prior to the GFC.

Additionally, we provide a categorical decomposition of these contributions by specific economic periods at horizon $h = 0$ (Table A2). We aggregate individual week-level contributions to the total LP-IV estimate and report the share of the absolute total contribution. This metric highlights which sub-periods exert the most significant influence on the "average" IRF and demonstrates that the stability of the aggregate estimate depends heavily on specific high-volatility regimes.

Table A1: High-Leverage Observations Across Currency Pairs (Horizon $h = 0$)

	USD/EUR		USD/GBP		EUR/GBP	
Rank	Date	c_{iv}	Date	c_{iv}	Date	c_{iv}
1	2008-06-06	-10.38	2008-08-15	13.26	2001-08-03	13.88
2	2008-07-04	8.59	2004-01-30	11.70	2003-02-07	12.77
3	2008-12-19	-7.75	2003-02-07	10.95	2001-05-11	-11.35
4	2007-12-14	7.59	2009-08-07	10.79	2000-06-09	9.87
5	2011-08-05	7.19	2007-09-21	-9.74	2001-11-09	9.71
6	2007-09-21	-6.97	2008-11-07	9.45	2007-01-12	-9.04
7	2011-03-04	-6.54	2003-05-09	-8.92	1999-09-10	-8.81
8	2008-09-19	6.25	2007-01-12	-8.81	2002-11-08	-8.19
9	2008-02-01	-6.06	2008-03-14	8.02	2008-11-07	7.38
10	2008-01-25	-5.80	2006-08-04	-7.97	2001-04-13	7.22

Note: The table reports the ten observations with the largest absolute contributions to the LP-IV estimate at horizon $h = 0$ for each currency pair. c_{iv} denotes the individual contribution of the observation to the aggregate impulse response.

Table A2: Decomposition of IRF Contributions by Economic Period (Horizon $h = 0$)

		USD/EUR		USD/GBP		EUR/GBP	
Period	N	$\sum c$	%	$\sum c$	%	$\sum c$	%
Pre-2007	408	-0.19	0.47	11.97	27.26	3.24	9.50
GFC (2007–2009)	156	-8.61	21.52	-6.75	15.37	-5.40	15.87
Eurozone Crisis (2010–2012)	157	9.96	24.91	-14.08	32.07	-11.87	34.85
2013–2020	365	-13.53	33.84	3.46	7.89	8.62	25.30
COVID (2020)	52	-2.79	6.97	-5.51	12.56	-2.75	8.08
Post-2020	254	4.91	12.28	2.13	4.86	-2.18	6.41
TOTAL	1392	-10.24	100.0	-8.78	100.0	-10.36	100.0

Note: N denotes the number of weekly observations. $\sum c$ is the sum of individual week-level contributions to the aggregate LP-IV estimate. % represents the share of the absolute total contribution, highlighting which periods exerted the most influence on the "average" IRF.

Table A3: Summary Statistics: Interest Rates and Monetary Shocks

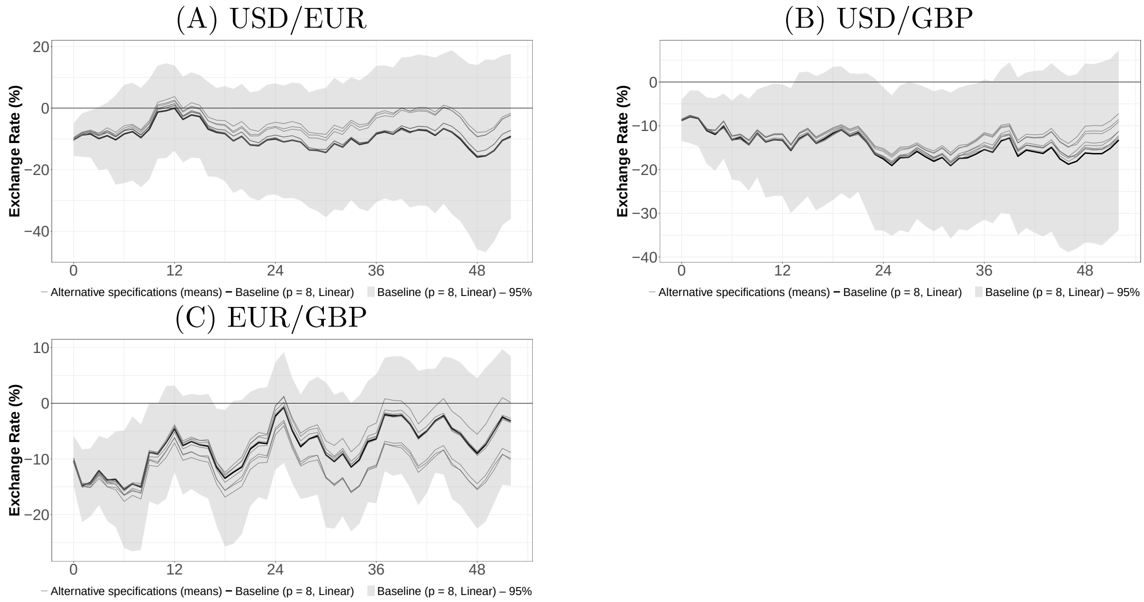
Variable	N	Mean	St. Dev.	Min	Max
USD/EUR					
Interest Rate Diff. Change	1,400	−0.024	8.514	−43.400	53.800
Monetary Shock Diff.	1,400	−0.191	3.180	−48.000	18.500
USD/GBP					
Interest Rate Diff. Change	1,400	0.031	8.945	−60.680	53.650
Monetary Shock Diff.	1,400	−0.002	4.073	−27.000	26.000
EUR/GBP					
Interest Rate Diff. Change	1,400	0.055	8.018	−63.580	57.800
Monetary Shock Diff.	1,400	0.244	3.600	−21.650	36.850

Note: Data is at a weekly frequency. Interest rates and shocks are in basis points.

B Specification Robustness: Lags and Trends

We examine the stability of our baseline results by varying the lag structure and the treatment of deterministic trends. Figure B1 displays the impulse responses across all possible combinations of lag lengths ($p \in \{4, 8, 12\}$) and trend specifications (none, linear, and quadratic). We choose $p = 8$ lags as our baseline (roughly two months of data at weekly frequency) to ensure that the information contained in prior ECB, Fed, and BoE meetings is fully absorbed by the controls. The estimated IRFs are broadly robust across these permutations. However, as noted in the decomposition analysis, this aggregate stability often masks substantial temporal heterogeneity in the underlying dynamics.

Figure B1: Robustness Check: Alternative Specifications



Note: IRF to a 100-bp increase in the one-year differential (black: mean; grey: Alternative specifications; shaded: 5–95% HAC CI). Alternative specifications are all other combinations of varying lags (4, 8, and 12) and trend specifications (none, linear, quadratic).

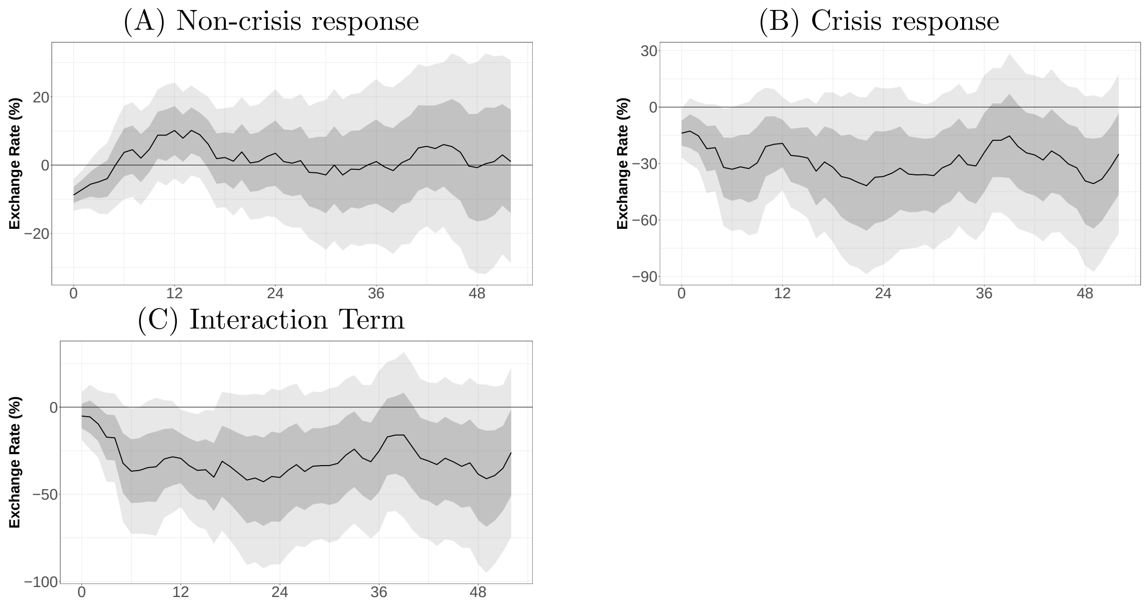
C State-Dependent LP-IV Specification

To formally test for regime-switching, we estimate a state-dependent Local Projection (LP-IV) model. We define a crisis dummy, I_t^C, based on NBER/CEPR recession dates and significant market dislocations. The dummy takes the value of 1 during the Global Financial Crisis (2007:M8–2009:M6), the Eurozone Sovereign Debt Crisis (2010:M1–2012:M12), and the onset of the COVID-19 pandemic (2020:M1–2020:M12). For each horizon h, we estimate the following second-stage regression:

$$e_{t+h} = \alpha_h + \beta_h \Delta d_t + \delta_h(\Delta d_t \times I_t^C) + \gamma_h(L)W_{t-1} + \text{trend}_t + \epsilon_{t+h} \tag{C1}$$

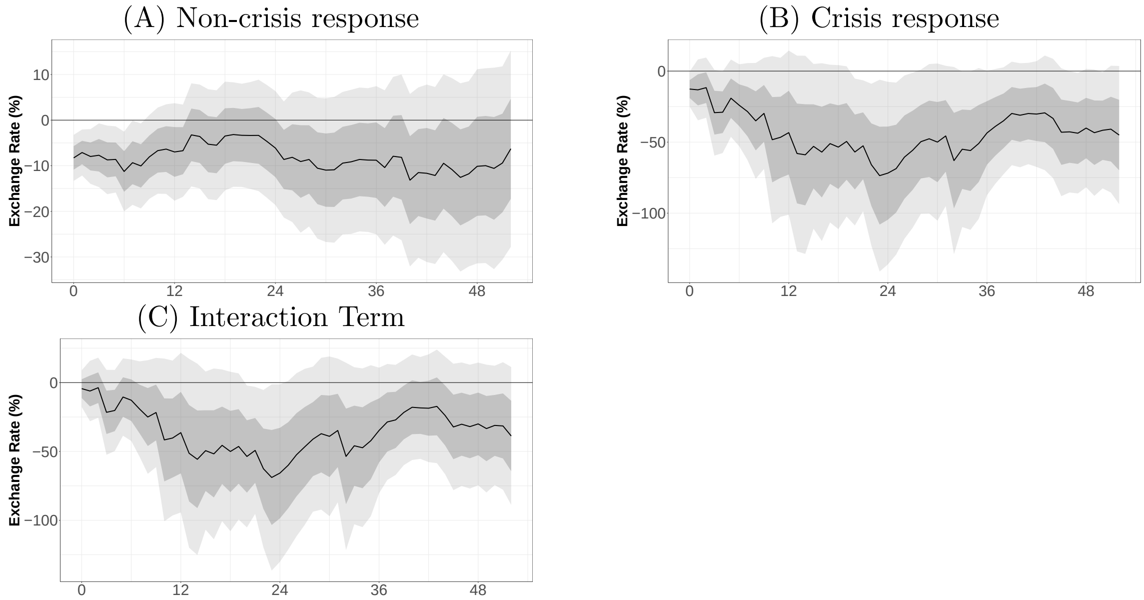
where e_{t+h} is the log exchange rate and Δd_t is the relative interest rate differential. W_{t-1} is a vector of p lags of the endogenous variables and exogenous controls. Inclusion of the interaction term introduces a second endogenous regressor. We use the high-frequency monetary surprise z_t and its interaction $(z_t \times I_t^C)$ as instruments for Δd_t and $(\Delta d_t \times I_t^C)$, respectively. This allows the first-stage pass-through and the second-stage exchange rate sensitivity to vary across regimes. Figures C1 through C3 report the "Non-crisis" response (β_h), the "Interaction" effect (δ_h), and the total "Crisis" response ($\beta_h + \delta_h$). The results strongly support the state-dependence hypothesis. While the non-crisis response exhibits some statistical significance on impact, its magnitude is substantially smaller. The pronounced appreciation and subsequent overshooting dynamics are captured overwhelmingly by the crisis-state parameters, with the non-crisis response attenuating rapidly and becoming statistically indistinguishable from zero across most subsequent horizons.

Figure C1: USD/EUR



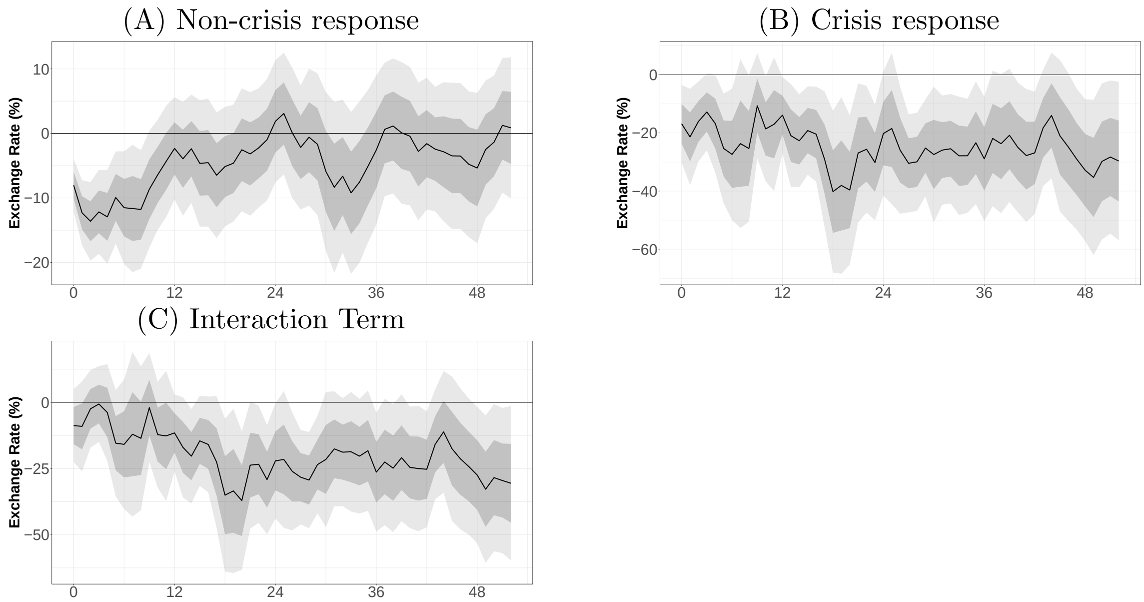
Note: IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% HAC CI).

Figure C2: USD/GBP



Note: IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% HAC CI).

Figure C3: EUR/GBP

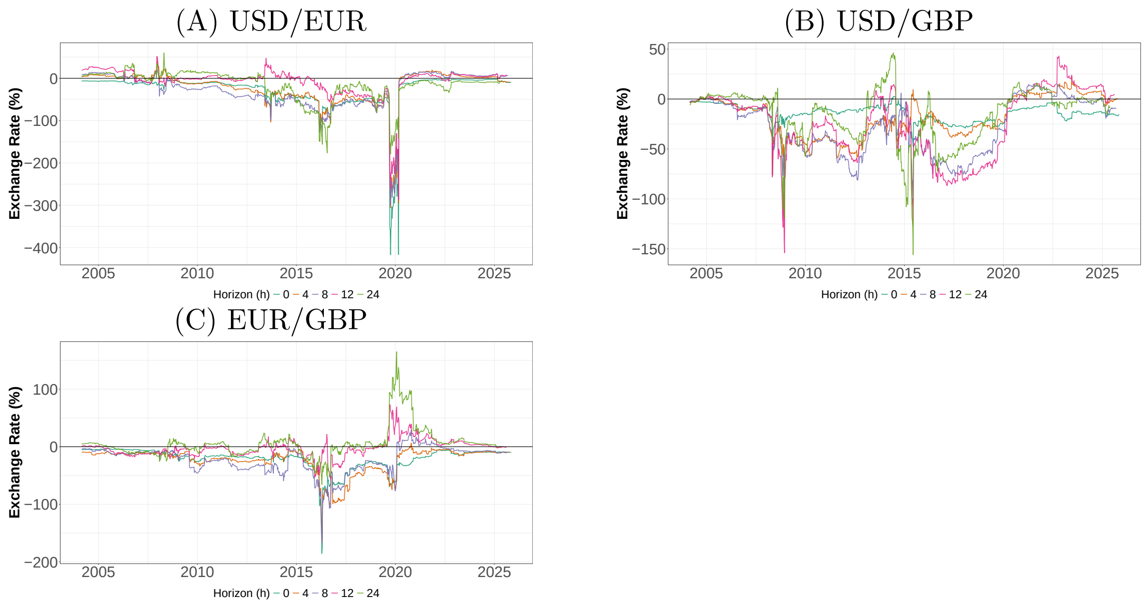


Note: IRF to a 100-bp increase in the one-year differential (black: mean; shaded: 16–84% / 5–95% HAC CI).

D Rolling Window Analysis

To ensure our findings are not an artifact of the exogenous definitions of crisis periods, we estimate the baseline LP-IV model over a rolling 5-year (260-week) window. Figure D1 plots the estimated coefficients over time for horizons $h = \{0, 4, 8, 12, 24\}$. For visual clarity, confidence intervals are omitted.The rolling estimates provide data-driven confirmation of state dependence. Across all horizons, the exchange rate sensitivity is most pronounced when the estimation window encompasses the GFC and Eurozone debt crisis. As these high-volatility weeks drop out of the rolling sample, the coefficients attenuate sharply toward zero. While this rolling approach is effectively a moving average of the influence-function decomposition, it confirms that the "average" IRF is not a structural constant but a time-varying parameter driven by specific historical dislocations.

Figure D1: Rolling Window LP-IV Analysis Across Horizons



Note: The figure reports the rolling LP-IV point estimates for a 5-year (260-week) window. The x-axis denotes the end-of-window date. Coefficients represent the response to a 100-bp increase in the one-year interest rate differential.